Hall Ticket Number:

MR-21

Code: B415CB3S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

Introduction to Database Management Systems

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write about data models?	2	1	1	1
1. b)	List the applications of Database systems?	2	2	1	1
1. c)	Discuss the use of Rename operation?	2	2	2	1
1. d)	Write about the basic form of SQL query?	2	2	2	1
1. e)	List out the properties of decomposition?	2	2	3	1
1. f)	What is redundancy and write about the problems with it?	2	1	3	2
1. g)	Write about serializability?	2	2	4	2
1. h)	Explain about transaction states?	2	2	4	2
1. i)	Write a short note on file organizations?	2	2	5	1
1. j)	Discuss about data on external storage?	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	What is E-R diagram and explain its Symbols?	5	3	1	3
2. b)	Explain the Architecture of the database?	5	4	1	1
	OR				
2. c)	Discuss about different types of integrity constraints?	5	3	1	3
2. d)	Write about the levels of abstraction in DBMS?	5	3	1	1
3. a)	Write about operations in relational algebra with an example?	5	3	2	1
3. b)	Discuss about Tuple relational calculus in detail?	5	3	2	1
	OR				
3. c)	Explain about NULL values and Views?	5	2	2	2
3. d)	Define trigger and explain all the three parts of it with an example?	5	2	2	2
4. a)	Discuss about trival and non trival functional dependencies?	5	4	3	2
4. b)	Illustrate the closure of attributes?	5	5	3	3
	OR				
4. c)	What is decomposition and explain its properties?	5	2	3	2
4. d)	Explain about First and Second normal forms	5	2	3	2
5. a)	List and explain ACID properties in detail?	5	2	4	1
5. b)	Explain the process of achieving implementation of atomicity and durability?	5	4	4	2
	OR				
5. c)	Discuss about Two-Phase locking protocol in detail?	5	2	4	2
5. d)	What is recovery and explain about log-based recovery with example?	5	2	4	2
6. a)	Explain about B+ tree indexing with an example?	5	2	5	2
6. b)	Explain the concept of cluster indexes with example	5	2	5	1
	OR				
6. c)	Explain about tree based indexing with an example?	5	2	5	1
6. d)	Explain in detail about ISAM?	5	2	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome